

Two-stage stochastic instances used in [Donnini et al. \(2025\)](#)

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In the following, we report the complete formulations of the instances considered in the computational test of [Donnini et al. \(2025\)](#).

1 Two-Stage Stochastic Linear Instances

1.1 Two-Stage Stochastic Knapsack Problem with Setup

The knapsack problem with setup [Michel et al. \(2009\)](#) is a two-stage variant of the knapsack problem that has pure binary first- and second-stage variables. In the knapsack problem with setup the set of items $J = \{1, \dots, n_y\}$ is partitioned into n_x classes $C_i \subseteq J$, $i \in I = \{1, \dots, n_x\}$ and the items in a class can be inserted into the knapsack only if the corresponding class was activated in the first stage. Activating a class involves a non-negative cost f_i and a non-negative reduction d_i of the knapsack capacity. Here we consider the stochastic version of this problem, in particular the cost q_{sj} of each item $j \in J$ is scenario dependent, as well as its weight w_{sj} and the maximum (c_s) and minimum (g_s) capacity of the knapsack. The objective is to choose a set of classes to be activated and a subset of items to be selected so that the difference between the average profit over all scenarios and the activation costs is maximized, while the total weight of the items lies in the prescribed interval between maximum and minimum capacity, computed by taking into account the class setup reductions. In particular, first stage binary variables x_i , $i \in I$, take value 1 if class i is activated, and second stage binary variables y_{sj} are set equal to one if item j is selected in scenario s . The extensive formulation of the problem is as follows:

$$\begin{aligned} \max \quad & - \sum_{i \in I} f_i x_i + \sum_{s \in S} p_s \left(\sum_{j \in J} q_{sj} y_{sj} \right) \\ \text{s.t.} \quad & \sum_{i \in I} d_i x_i + \sum_{j \in J} w_{sj} y_{sj} \leq c_s, & s \in S, \\ & \sum_{i \in I} d_i x_i + \sum_{j \in J} w_{sj} y_{sj} \geq g_s, & s \in S, \\ & y_{sj} \leq x_i, & i \in I, j \in C_i, s \in S, \\ & x_i \in \{0, 1\}, & i \in I, \\ & y_{sj} \in \{0, 1\}, & j \in J, s \in S. \end{aligned}$$

1.2 Two-Stage Stochastic Capacitated Facility Location Problem

For the capacitated facility location problem we considered the stochastic version of the classical formulation from the OR library <http://old.math.nsc.ru/AP/benchmarks/CFLP/cflp-eng.html>, with the addition of a new set of constraints. In particular, for each facility location $i \in I = \{1, \dots, n_x\}$, we have an opening cost c_i , and a maximum and minimum value of the production that must be carried out if that facility is opened, respectively V_i and v_i . For every pair of facility i and client $j \in J = \{1, \dots, n_y\}$, there is a non-negative cost g_{sij} if client j is served by facility i , and a non-negative demand d_{sij} that

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client j requires from facility i , in scenario $s \in S$. Binary variables x_i take value 1 if facility i is opened, while second stage binary variable y_{sij} is set to 1 if client j is served by facility i in scenario s . The extensive formulation of the problem is the following:

$$\begin{aligned}
\min \quad & \sum_{i \in I} c_i x_i + \sum_{s \in S} p_s \left(\sum_{i \in I} \sum_{j \in J} g_{sij} y_{sij} \right) \\
\text{s.t.} \quad & \sum_{i \in I} y_{sij} = 1, & j \in J, s \in S, \\
& \sum_{j \in J} d_{sij} y_{sij} \leq V_i x_i, & i \in I, s \in S, \\
& \sum_{j \in J} d_{sij} y_{sij} \geq v_i x_i, & i \in I, s \in S, \\
& x_i \in \{0, 1\}, & i \in I, \\
& y_{sij} \in \{0, 1\}, & i \in I, j \in J, s \in S.
\end{aligned}$$

2 Two-Stage Stochastic Quadratic Instances

2.1 Two-Stage Stochastic Quadratic Knapsack Problem with Setup

The quadratic knapsack problem with setup [Galli et al. \(2025\)](#), is the generalization of the previous problem that includes a quadratic objective function. It inherits all parameters and constraints from the knapsack problem with setup, with the addition of pairwise profits $h_{sjj'}$ which contribute to the objective function if both items j and j' are inserted into the knapsack in scenario s . Once again, we considered the stochastic version of the problem and the extensive formulation can be written as follows:

$$\begin{aligned}
\max \quad & - \sum_{i \in I} f_i x_i + \sum_{s \in S} p_s \left(\sum_{j \in J} q_{sj} y_{sj} + \sum_{\substack{j \in J, j' \in J: \\ j < j'}} h_{sjj'} y_{sj} y_{sj'} \right) \\
\text{s.t.} \quad & \sum_{i \in I} d_i x_i + \sum_{j \in J} w_{sj} y_{sj} \leq c_s, & s \in S, \\
& \sum_{i \in I} d_i x_i + \sum_{j \in J} w_{sj} y_{sj} \geq g_s, & s \in S, \\
& y_{sj} \leq x_i, & i \in I, j \in C_i, s \in S, \\
& x_i \in \{0, 1\}, & i \in I, \\
& y_{sj} \in \{0, 1\}, & j \in J, s \in S.
\end{aligned}$$

2.2 Two-Stage Stochastic Quadratic Capacitated Facility Location Problem

The quadratic capacitated facility location problem is a direct generalization of the previous problem. In fact, it inherits all of its parameters and constraints, and the only addition is represented by pairwise costs h_{si} which are included in the objective function if more than one client is served by facility i in

scenario s . The extensive formulation of the problem is as follows:

$$\begin{aligned}
\min \quad & \sum_{i \in I} c_i x_i + \sum_{s \in S} p_s \left(\sum_{i \in I} \sum_{j \in J} g_{sij} y_{sij} + \right. \\
& \left. \sum_{i \in I} h_{si} \left(\sum_{j \in J} \sum_{j' \in J: j < j'} y_{sij} y_{sij'} \right) \right) \\
\text{s.t.} \quad & \sum_{i \in I} y_{sij} = 1, & j \in J, \ s \in S, \\
& \sum_{j \in J} d_{sij} y_{sij} \leq V_i x_i, & i \in I, \ s \in S, \\
& \sum_{j \in J} d_{sij} y_{sij} \geq v_i x_i, & i \in I, \ s \in S, \\
& x_i \in \{0, 1\}, & i \in I, \\
& y_{sij} \in \{0, 1\}, & i \in I, \ j \in J, \ s \in S.
\end{aligned}$$

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